

DigiPermit — Final Project Report

Module Integration: Development Software 3 (DS3) × Information Systems 3 (IS3) × Internet Programming 2 (IP2)

Project Title: DigiPermit: A Multi-User Foreign-National Visa and Permit Compliance Monitoring System

Duration: 6 Weeks (4 Sprints)

Methodology: Agile (Scrum)

Declaration

This report describes the design, implementation, and evaluation of DigiPermit. AI-assisted tools (including Cursor/ChatGPT) were used for code generation and documentation drafting. All team members must be able to explain design decisions and demonstrate understanding of the implemented system during the external moderation.

1. Problem Statement & Strategic Context

Foreign nationals holding work visas, study visas, and other immigration-related documents must comply with expiry dates and renewal conditions. Employers, universities, and clinics often track these records manually using spreadsheets and email, leading to:

- Missed expiry dates and delayed renewals
- No real-time verification at checkpoints
- No audit trail for compliance decisions
- Unauthorised access to sensitive records

Strategic significance: Non-compliance creates legal, operational, and reputational risk for organisations employing or enrolling foreign nationals.

Limitation: DigiPermit is a compliance-monitoring platform. It does not issue official visas or replace the Department of Home Affairs or any official immigration authority.

2. Target Users & Stakeholders

Stakeholder	Role in system
System Administrator	Platform governance
Employer HR Officer	Employee work-visa compliance
University International Office	Student study-visa monitoring
Clinic Administrator	Patient/employee document records
Foreign National	Self-service permit visibility
Verification Officer	Checkpoint verification
Immigration Officer (simulated)	Record validation
Manager/Auditor	Read-only analytics

3. System Objectives

1. Centralise permit compliance records in a secure multi-tenant database
2. Automate expiry monitoring and notifications
3. Provide multi-channel verification (manual, QR, RFID simulation)
4. Detect suspicious verification activity using rule-based analytics
5. Deliver role-specific dashboards and Power BI executive reporting
6. Demonstrate full-stack integration across IS3, IP2, and DS3 outcomes

4. Functional Requirements (Summary)

- Authentication and role-based access control (8 roles)
- CRUD for organisations, users, foreign nationals, permits, permit types
- Verification with ten result states and audit logging
- Automated daily expiry job with notifications at 90→0 days
- IoT scan simulation feeding database in real time
- Renewal/update request workflow
- Analytics API and SQL views for Power BI
- AI-assisted executive insight summaries

Full requirements: see `docs/PHASE1-PLANNING-AND-DESIGN.md`.

5. Non-Functional Requirements

Category	Implementation
Security	JWT auth, RBAC middleware, RLS, passport masking, <code>.env</code> secrets
Performance	Indexed queries on <code>permit_number</code> , <code>expiry_date</code> , <code>organisation_id</code>
Usability	Ionic mobile-first UI, loading states, status badges
Maintainability	MVC backend, versioned SQL migrations
Auditability	Immutable verification logs, soft archival

6. System Design

6.1 Architecture (Three-tier)

Presentation: Ionic Angular (TypeScript)
Business Logic: Node.js + Express (REST API)
Data: Supabase PostgreSQL + Supabase Auth

6.2 Design artefacts (IS3)

Artefact	Location
BPMN As-Is / To-Be	<code>docs/IS3-BPMN-PROCESS-MODELS.md</code>
3NF ERD	<code>docs/IS3-ERD.md</code>
IoT Connectivity Map	<code>docs/IS3-IOT-CONNECTIVITY-MAP.md</code>
AI Workflow	<code>docs/IS3-AI-WORKFLOW.md</code>

6.3 Database

- 12 normalised tables with foreign keys and check constraints
- 7 analytics views for Power BI
- Row Level Security for organisation isolation
- Live IoT data in `iot_scan_events`

7. Implementation

7.1 Backend (IP2)

- **Stack:** Node.js, Express.js, `@supabase/supabase-js`
- **Pattern:** Routes → Controllers → Services → Supabase
- **Endpoints:** 40+ REST routes documented in Postman
- **Validation:** express-validator pattern, global error handler
- **Jobs:** `expiryCheckJob.js` (node-cron, daily 06:00 UTC)

7.2 Frontend (IP2)

- **Stack:** Ionic 8, Angular 20, TypeScript
- **Features:** Route guards, auth interceptor, role-based sidebar
- **Pages:** 8 role modules with dashboards and CRUD forms

7.3 IoT Integration (IS3)

Simulated RFID/QR devices POST to `/api/iot/simulate`. Backend validates permit, stores `iot_scan_events` and `verification_logs`, creates alerts when required. Cisco Packet Tracer topology documented for presentation.

7.4 AI & Analytics (IS3)

- Rule-based suspicious-activity detection in verification flow
- `GET /api/analytics/ai-insights` — executive summary generation
- Power BI connects to Supabase views (`vw_expiring_permits`, etc.)

8. Agile Process (DS3)

Sprint	Week	Deliverable
Sprint 1	2	Backend + database foundation
Sprint 2	3	Full API, IoT, HR frontend
Sprint 3	4	All roles UI, analytics, AI
Sprint 4	5	Testing, report, presentation

Sprint reviews: `docs/sprints/SPRINT-*-REVIEW.md`

Product backlog: `docs/PHASE1-PLANNING-AND-DESIGN.md` Section 13

9. Testing

Test plan: docs/TEST-PLAN.md

Postman collection: digipermit-backend/postman/DigiPermit-API.postman_collection.json

Sample results: - Authentication: PASS - CRUD operations: PASS - Verification (valid/expired/revoked): PASS - IoT simulation: PASS - Role isolation: PASS

10. Demonstration Scenario

See docs/DEMO-SCRIPT.md for the full 23-step flow.

Summary: HR registers employee → captures permit → employee views QR → verification officer scans → admin views alerts → Power BI dashboard.

11. Challenges & Lessons Learned

Challenge	Resolution
Supabase CLI unavailable on Windows	Used SQL Editor + combined setup-all.sql
CORS port mismatch (4200 vs 8100)	Multi-origin CORS configuration
Academic scope vs feature breadth	Prioritised 4 core roles first, extended to 8
IoT hardware unavailable	HTTP simulation via API + Packet Tracer map

12. Conclusion

DigiPermit successfully integrates backend development (IP2), digital firm architecture pillars (IS3), and Agile software engineering practices (DS3) into a single working system. The platform automates the manual expiry-tracking bottleneck identified in the As-Is BPMN model, feeds simulated IoT data into a 3NF database, and supports executive decision-making through Power BI and AI insight summaries.

References

- Ionic Framework Documentation — <https://ionicframework.com/docs>
- Supabase Documentation — <https://supabase.com/docs>
- OMG BPMN 2.0 Specification

- Express.js Guide — <https://expressjs.com>
 - Microsoft Power BI Documentation
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Appendices

Appendix	File
A. API Specification	<code>docs/API-SPECIFICATION.md</code>
B. Demo Script	<code>docs/DEMO-SCRIPT.md</code>
C. Power BI Setup	<code>docs/POWER-BI-SETUP.md</code>
D. Test Plan	<code>docs/TEST-PLAN.md</code>
E. Presentation Slides	<code>docs/PRESENTATION-SLIDE-DECK.md</code>

Report length: ~12 pages (when exported to PDF from this document)